QUANTUM IAS & PCS

TATTVA SERIES

UPPSC PRELIMS | FAST REVISION

Indian Geography

PYQ-Based Revision Pointers

Sample Chapter

Every pointer comes from an actual UPPSC question · Frequency marks · Year tags
Asked-Most Boxes · Trap Alerts · Quick Tables · Last-Minute 10

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Chapter 6

Drainage I: The Himalayan Rivers

Built from 36 UPPSC questions (1990–2025). Focus: the Ganga system, Satluj, Bundelkhand rivers, Brahmaputra and Indus.

ASKED MOST · REVISE FIRST

- **Satluj** is the most-asked river: rises at **Rakas Tal** (Tibet), enters India at **Shipki La**, cuts across **all Himalayan ranges**, is **trans-Himalayan** and **does not rise in India** 6× 2010, 2012, 2013, 2014, 2020, 2023
- **Devprayag** = **Alaknanda + Bhagirathi**; the river is called **Ganga** from here 4× 2002, 2016, 2023, 2024
- **Betwa** (Bundelkhand): joins the **Yamuna**; **Rajghat dam** is on it; **Ken-Betwa** (UP-MP) is India's first river link 4× 2006, 2006, 2014, 2015
- Himalayan rivers are **perennial** because of **snow and glacier melt**, not the monsoon (Assertion-Reason) 2× 2011, 2025

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

6.1 Nature of Himalayan Rivers

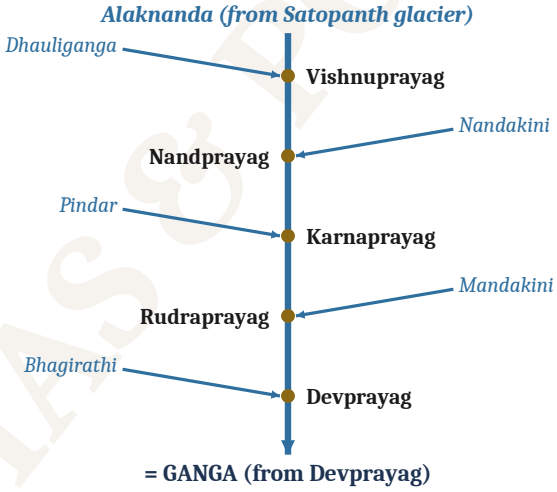
- Himalayan rivers are **perennial**: summer **snow-melt** keeps them flowing; the monsoon explains only the rainy-season flood 2× 2011, 2025
- **Ganga** is an **antecedent** river: older than the mountains, it kept cutting down as they rose 1× 2013
- **Chambal** is a **superimposed** river (also Son, Banas, Subarnarekha) 1× 2007
- **Satluj** cuts across **all ranges** of the Himalaya (Zaskar, Greater, Lesser, Shiwalik) 2× 2012, 2014
- **Trans-Himalayan** rivers (rise north of the Himalaya): **Satluj**, Indus, Brahmaputra 1× 2020

Drainage type	Key idea	Example
Antecedent	Older than the mountain	Ganga, Indus, Satluj, Brahmaputra
Consequent	Follows the original slope	Godavari, Krishna, Kaveri
Subsequent	Along weak rock bands	Strike-valley tributaries
Superimposed	Let down onto older rocks	Chambal, Son, Banas

6.2 The Ganga and its Headwaters

- Ganga rises as the **Bhagirathi** at **Gaumukh** (Gangotri glacier); called **Ganga** from **Devprayag** 4× 2002, 2016, 2023, 2024
- **Badrinath** is on the **Alaknanda**; **Kedarnath** on the **Mandakini** 2× 2002, 2013
- **Rudraprayag** = Alaknanda + **Mandakini**; **Karnaprayag** = Alaknanda + **Pindar**; **Tehri** = Bhagirathi + **Bhilangna** 2× 2002, 2024
- Adi Shankaracharya's four mathas: **Jyotirmath, Sringeri, Dwarka, Puri**; **Kedarnath is not** one of them 1× 2002
- In Bangladesh the Ganga is the **Padma**; the Bhagirathi-Hooghly branch flows past Kolkata 1× 2016
- Ganga pollution comes from **sewage and industrial waste** (Kanpur tanneries); "holier = more polluted" is false 1× 2005

PANCH PRAYAG: WHERE THE ALAKNANDA'S TRIBUTARIES JOIN

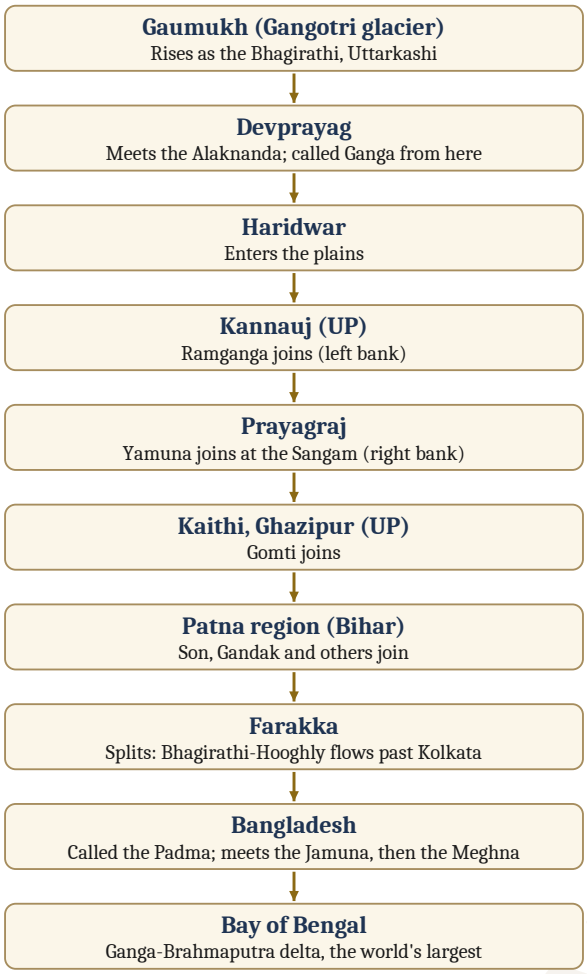


TRAP ALERT

- **Devprayag** (Bhagirathi) and **Rudraprayag** (Mandakini) are the usual swap. Tehri is **Bhagirathi + Bhilangna**, not an Alaknanda confluence.

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COURSE OF THE GANGA, SOURCE TO SEA

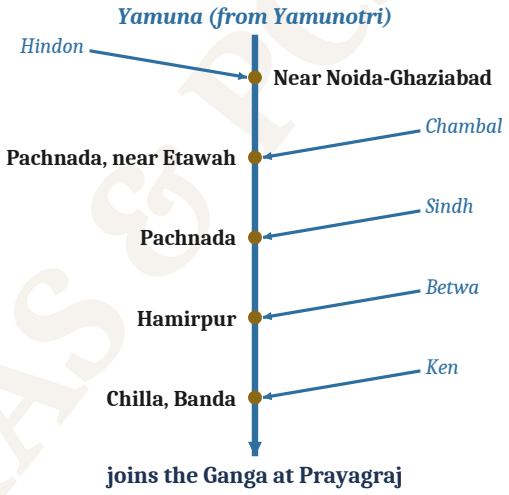


6.3 Tributaries of the Ganga

- **Gomti** is the only Ganga tributary that **rises in the plains**: Phulhar Tal, **Pilibhit**; flows past Lucknow and Jaunpur 1× 2008
- **Yamuna** is **perennial**; it is "dead" below Delhi because water is diverted and sewage added (A true, R false) 1× 2015
- **Ramganga** joins the **Ganga** (at Kannauj), **not** the Yamuna 1× 2010
- Yamuna's right-bank tributaries, **west to east**: **Chambal, Sindh, Betwa, Ken** 1× 2020
- **Son** has the largest catchment among Son, Gandak, Ramganga and Mahananda (about 70,000 sq km) 1× 2017
- **Gandak** does **not** flow through UP; it flows through Bihar 1× 2021
- **Jonk** belongs to the **Mahanadi** basin, not the Ganga 1× 2021
- **Kosi**, the "Sorrow of Bihar", **shifts its course** the most 1× 2013
- **Mahatma Gandhi Setu** (Patna-Hajipur) is in **Bihar** 1× 1993
- **Moribund delta** = western part of the **Bengal (Ganga-Brahmaputra) delta** 1× 2019
- Tributary match: **Ganga-Son, Godavari-Manjira, Krishna-Bhima, Yamuna-Ken** 1× 2003

Tributary	Bank	Joins at
Ramganga	Left	Kannauj (UP)
Gomti	Left	Kaithi, Ghazipur (UP)
Ghaghara	Left	Near Chhapra (Bihar)
Gandak	Left	Sonepur-Hajipur (Bihar)
Kosi	Left	Kursela (Bihar)
Yamuna	Right	Prayagraj (Sangam)
Son	Right	Near Patna

YAMUNA'S TRIBUTARIES, WEST TO EAST (ASKED 2020)



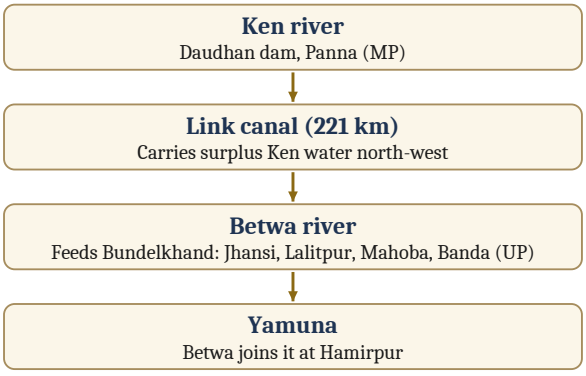
UP LINK

Gomti (Pilibhit to Lucknow to Jaunpur to Kaithi), the Yamuna's Bundelkhand tributaries (Chambal near Etawah, Betwa at **Hamirpur**, Ken at Chilla in **Banda**) and the Kanpur-Prayagraj-Varanasi stretch of the Ganga are UP's most-asked river facts.

6.4 Ken-Betwa and Bundelkhand

- **Ken-Betwa Link**: **UP and MP**, MoU **2005**; India's **first** river-interlinking project; foundation stone **25 Dec 2024** 1× 2006
- **Betwa** joins the **Yamuna** (at Hamirpur) 1× 2014
- **Rajghat dam** (Lalitpur, joint UP-MP project) is on the **Betwa** 2× 2006, 2015

KEN-BETWA LINK: DIRECTION OF WATER



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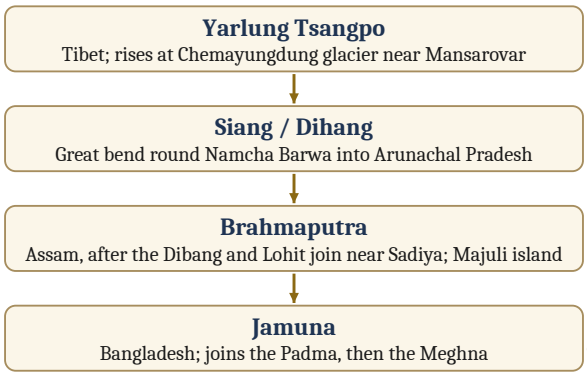
TRAP ALERT

- Water goes **from the Ken to the Betwa** (Daudhan dam on the Ken). Both are tributaries of the **Yamuna**, not the Ganga.

6.5 The Brahmaputra System

- **Indus and Brahmaputra** rise at almost the same point, near **Mansarovar-Kailash** 1× 2006
- Brahmaputra basin: **Tibet, India, Bangladesh** (and Bhutan) 1× 1990
- **Teesta** was once a Ganga tributary; after the **1787 flood** it joined the **Brahmaputra**: **river capture** (A and R true, R explains A) 1× 2021
- **Sankosh** forms the **Assam-West Bengal** boundary 1× 2007

ONE RIVER, FOUR NAMES: THE BRAHMAPUTRA



6.6 The Indus System

- **Satluj** rises at **Mansarovar-Rakas Tal** and enters India at **Shipki La** 1× 2023
- **Satluj does not rise in India**; Beas, Ravi and Chenab do 2× 2010, 2013
- **Kishanganga** (Neelum) is a tributary of the **Jhelum** 1× 2013
- **Doodh Ganga** flows in **Jammu and Kashmir** (also a stream of that name in Uttarakhand) 1× 2015
- World's **highest railway bridge**: on the **Chenab** (Reasi, 359 m; opened June 2025) 1× 2006

River	Source
Jhelum	Verinag spring
Chenab	Chandra + Bhaga at Tandi
Ravi	Near Rohtang
Beas	Beas Kund
Satluj	Rakas Tal (Tibet)

LAST-MINUTE 10

1. Perennial Himalayan rivers = **snow-melt**
2. Ganga = **antecedent**; Chambal = **superimposed**
3. **Satluj**: Rakas Tal, Shipki La, cuts all ranges, trans-Himalayan
4. **Devprayag** = Alaknanda + Bhagirathi
5. **Badrinath** on Alaknanda; **Kedarnath** on Mandakini
6. **Gomti** = only plains-born tributary (Pilibhit)
7. Yamuna tributaries west to east: **Chambal, Sindh, Betwa, Ken**
8. **Ken to Betwa** link, UP-MP; Rajghat dam on Betwa
9. Teesta = **river capture** (1787); Kosi = Sorrow of Bihar
10. Kishanganga to **Jhelum**; Chenab bridge = world's highest

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